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TABLETS WITH SPORES USED IN ABSOLUTE POLLEN ANALYSIS

BY

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SOMMAIRE. — Des comprimés de calcium contenant un nombre déterminé de spores de *Lycopodium* sont additionnés à des sédiments. Lors d'une analyse partielle il est possible d'évaluer la somme totale des pollens contenus dans un échantillon à l'aide du nombre de spores de *Lycopodium* observées. Le conditionnement et la calibration des comprimés, leur utilisation sont expliqués et les possibilités d'erreurs statistiques sont évaluées.

Introduction.

Methods for absolute pollen determination were described by BENNINGHOFF 1962, DAVIS 1965 and JØRGENSEN 1967.

BENNINGHOFF (l. c.) adds a measured amount of exotic pollen suspended in glycerine. When the pollen concentration in the added suspension is known, and the exotic pollen grains are counted like other pollen-types it is possible to calculate the total number of pollen grains in the pollen sample. The advantage of using this method is that it is not necessary to count all the material in a slide and that loss of a small part of the sample is immaterial when exotic pollen and pollen sample are mixed. Furthermore no measuring except of the added suspension and of the pollen sample are necessary. However glycerine as an embedding medium is not suitable as it may evaporate or absorb water. Other liquids may be used, but in all cases, if the method is meant to be an advantage, the suspension has to be rather large. But it is difficult to homogenize a large suspension every time it has to be used.

DAVIS' (l. c.) method is based on total analysis of volume aliquots from a prepared sample. However, it is difficult to take volume aliquots which have the same concentration as the total sample. Fur-

thermore the accuracy of volume aliquots is not as large as the accuracy of weighed aliquots as used by JÖRGENSEN (1. c.). JÖRGENSEN's method is more accurate but both methods can not take the loss of pollen grains during the preparation into account.

Hence it was decided to produce a suspension in a solid matter which is nonhygroscopic and easily soluble. When a powder and exotic pollen or spores are homogenized, and this matter is made solid, no later homogenizing is necessary. If the solid matter is divided in equal small aliquots it is not necessary to weigh the added suspension.

An attempt has been made to find an artificial product which could replace exotic pollen or spores. This did not succeed as it was not possible to find a material which has the same chemical and physical properties.

THE TABLETS.

Tablets made of calcium carbonate soluble in HCl were preferred.

Spores of *Lycopodium clavatum* L. were used because they were easily available and because they were of a reasonable size and easy to recognize. The tablets are not usable in samples which contain spores of *Lycopodium clavatum* (1). The tablets were produced by the formulation department of a drug firm as follows :

2 g *Lycopodium* spores
1000 g precipitated calcium carbonate
700 g lactose
9 g Prejel PA5

were mixed and granulated with

350 g distilled water with 4 % Carbowax 4000.

Prejel PA5 (water soluble polymeric starch) is a tablet dispersing agent, which gave some precipitate with HCl and alcohol. If the preparation of pollen samples starts with HCl it may be superfluous.

Carbowax 4000 (polyethylene glycol from Union Carbide) is an excipient and glazing agent.

Granulation is a process that compounds the mixed powder in small lumps in order that the material will not stick in the tablet machine.

After drying

9 g Carbowax 4000
73 g lactose

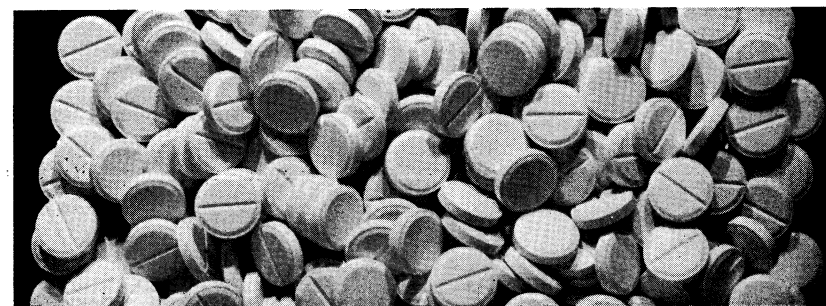
were added. Carbowax is used for glazing and lactose for making the tablets harden.

(1) This problem, however, is solved in the postscript.

180 mg-tablets were produced at a machine pressure of 5.5 kg per cm².

About 20 000 spores per tablet were expected.

The tablets are slightly hygroscopic. They were stored at a rather constant temperature and humidity and no changes in weight have been observed during half a year.



The tablets in natural size.

One hundred tablets were weighed with an accuracy of 10⁻⁵ grams each. Weight distribution, arithmetic mean, standard deviation and coefficient of variation are shown in fig. 1.

The extremely high and low values in fig. 1 represent tablets which were visibly thicker or thinner than the others.

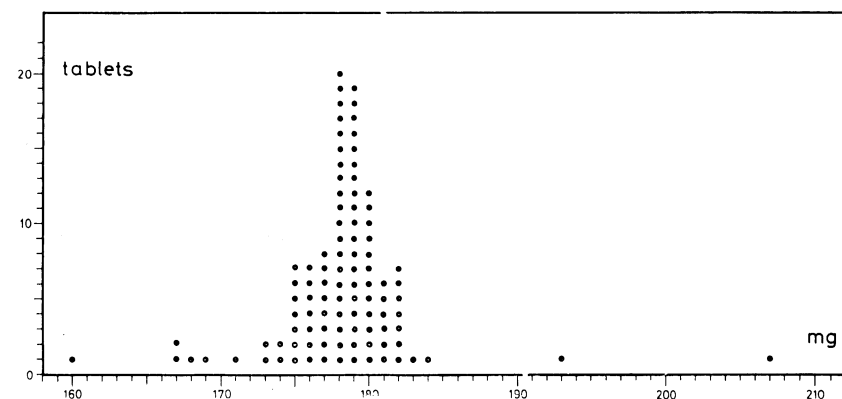


Fig. 1. — Weight distribution of single tablets.

Class intervals of 1 mg.

Arithmetic mean, $\bar{X} = \frac{\sum X}{N} = 0.17819$ g.

Standard deviation, $s = \pm \sqrt{\frac{(\bar{X} - X)^2}{N}} = \pm 0.00480$ g.

Coefficient of variations, $V = \frac{s}{\bar{X}} 100 = \pm 2.7$ per cent

The large range of variation is presumably due to the fact that magnesium stearate and talc powder which are components in the ordinary tablet production were omitted because they gave problems in dissolving. If they were used the tablets would have had a more smooth surface.

As 5 tablets are expected to be a convenient dosage 33 samples of 5 tablets each were weighed (fig. 2).

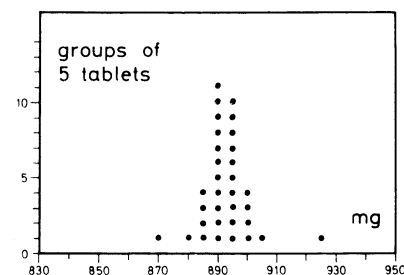


FIG. 2. — Weight distribution of groups of 5 tablets.

Class intervals of 5 mg.

$\bar{X} = 0.89258$ g

$s = \pm 0.00872$ g

$V = \pm 1.0$ per cent

Determination of the spore concentration.

The chemical preparation demands great accuracy and care so that no material is lost.

The centrifugates were agitated only with a Whirlimixer (Scientific Industries, U.K.) so as to prevent loss of material.

The liquids were removed from the centrifuge tubes with a pipette except for the last ml.

1) 5 tablets were weighed and placed in a weighed glass centrifuge tube.

A small amount of crushed diatoms was added to promote settling of the spores during centrifugations.

About 2 ml. alcohol were added.

40 % HCl sufficient to dissolve the tablets was added. Foam was kept down with alcohol.

More alcohol was added and the material centrifuged.

2) The material was washed twice with 96 % alcohol and absolute alcohol, and once with xylene and benzene.

Staining with fuchsin was done during the last washing with absolute alcohol.

After the last centrifugation the material was not stirred but 10 drops of silicone oil were added and the benzene evaporated for two days.

3) The centrifuge tube was weighed and the weight of the silicone oil suspension calculated.

After 10-15 minutes stirring with a spatula aliquots were placed on weighed slides. The slides with the aliquots were weighed and the weight of aliquots were calculated.

The total number of spores in the aliquots was counted and the total spore contents in 5 tablets calculated.

5 samples of 5 tablets each were treated in this way with the following result :

TABLE I.

Sample No	Weight of			Number of spores		
	5 tabl. (g)	Silicone oil suspension (g)	Aliquots (g)	in aliquots	in 5 tabl.	per g tabl.
1	0.89176	0.44439	0.02787	6608	105.365	118.154
2	0.89022	0.42466	0.02449	5975	103.607	116.384
3	0.89132	0.44385	0.02260	5614	110.255	123.699
4	0.89238	0.44242	0.02285	5289	102.402	114.755
5	0.89278	0.45520	0.01698	3860	103.479	115.906
$\bar{X} = 0.89169$				$\bar{X} =$	105.022	117.780
				$\hat{\sigma} =$	± 3112	± 3528
				$\frac{\hat{\sigma}}{\bar{X}} \cdot 100 =$	$\pm 3.0\%$	$\pm 3.0\%$

The estimated standard error, $\hat{\sigma}$ is calculated from the equation $\hat{\sigma} = \pm \sqrt{\frac{(\bar{X} - X)^2}{N - 1}} \cdot \frac{\hat{\sigma}}{\bar{X}} \cdot 100$ is the estimated standard error expressed in per cent of the mean, $\bar{X}$.

ABSOLUTE POLLEN ANALYSIS.

To a measured (weight, volume) pollen sample is added e.g. 5 tablets before the preparation. The preparation is carried out in the usual way starting with either KOH or HCl. When starting with HCl it might be preferred to add a small amount of alcohol in advance and then use 40 % HCl. When the tablets are dissolved and the sample homogeneously mixed, loss of a small part of the sample is immaterial.

Analysis is carried out as in ordinary pollen analysis and *Lycopodium* spores counted like other pollen and spore types. This is done best by a foot counter. The number of pollen grains in the sample is found from the equation :

$$\frac{105\ 000 \times \text{number of pollen grains counted}}{\text{number of } Lycopodium \text{ spores counted}} = \text{total number of pollen grains in a sample.}$$

ERRORS.

The accuracy of the result depends on the numbers of grains counted.

The error from the tablet calibration is 3 % (table 1). To this should be added the errors on the spore and the pollen counts which equal the square roots of the numbers counted. Hence the total error $\frac{\sigma_1 \times 2}{3}$ on the analysis can be found as

$$\frac{\sigma_1 \times 2}{3} \% = \pm 100 \sqrt{\left(\frac{\sigma_1}{r_1}\right)^2 + \left(\frac{\sigma_2}{r_2}\right)^2 + \left(\frac{\sigma_3}{r_3}\right)^2}$$

σ_1 is the standard error on r_1 which is the number of spores in 5 tablets. $\sigma_1 \sim \hat{\sigma}$ and $r_1 \sim \bar{X}$ in table I.

σ_2 is the standard error on r_2 , which is the number of pollen grains counted in the sample, $\sigma_2 = \pm \sqrt{r_2}$

σ_3 is the standard error on r_3 , which is the number of *Lycopodium* spores counted in the sample $\sigma_3 = \pm \sqrt{r_3}$

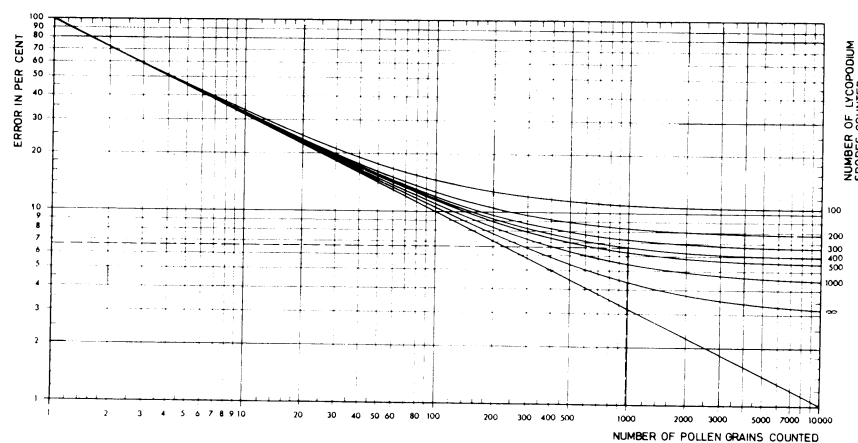


Fig. 3. — The estimated error on the calculated number of pollen grains in a sample when tablets are used. Error in per cent. The straight line is the error on the pollen counts alone. The curves show the total error when the error on the pollen counts, on the tablets and on the *Lycopodium* spores counted are taken into account. The dashed lines refer to the example mentioned in the text.

From this equation fig. 3 is constructed. It is drawn on double logarithmic paper. The upper curve shows the error when 100 spores are counted, the next when 200 are counted, etc. The straight line shows the error on the pollen counts alone. This error is found in all pollen analysis.

So it is seen that when e.g. 400 *Lycopodium* spores are counted the error on 100 pollen grains counted will only increase from 10 to

about 12 per cent due to the use of tablets. But when 1000 pollen grains are counted the error will increase from about 3 to almost 7 per cent.

This means that the more pollen grains are counted the greater part of the error is due to the use of tablets, although the total error decreases.

EXAMPLE.

Five tablets were added to a sample from an Atlantic varved gyttja. 998 fossil pollen grains and 400 *Lycopodium* spores were counted. The total pollen content is $\frac{105\ 000}{400} \times 998 = 261\ 975$. After adjustment for volume and number of varves a pollen sedimentation of 12 950 grains per year per cm^2 was found. The standard error is found from fig. 3 (the dashed lines) to be $\pm 6.6\ %$ equal to ± 850 grains per year per cm^2 .

Postscript.

A new production of tablets has been made for distribution. The spores in these are dark brownish coloured by acetolysis. So naturally occurring *Lycopodium* should not be lost. The new tablets are dissolved in KOH before treatment with HCl to prevent formation of sparingly soluble organic salts. The spore content was determined by an electronic blood cell counter.

Summary.

Calcium tablets with known content of *Lycopodium* spores are used as an additive to the pollen samples. When the spores are counted during the pollen analysis it is possible to calculate the total pollen content. Tablet production, calibration and use is described and the statistical errors are calculated.

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